(a) From trained models to the exact split

Models, CIFAR-10

- **V-scr** 3 runs
- **V-IN** 2 runs
- **R224-scr** 2 runs
- **R224-IN** 2 runs
- **R32** ResNet-18, 3 runs

seed pair with V-scr;
CIFAR-100: 3 + 3 runs

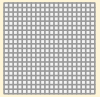
Fixed panels

- **C10** 7 + 1
- **C100** 7 + 2
- **IN-V2** 6 + 1

robust CNNs + secondary; IN-V2: 3 pairs

PGD-10 →

Transfer matrix T



20 / 15 / 14 models, target-correct

per pair →

Exact split

log T = log w + log τ + ½ r

w vulnerability, τ symmetric similarity, r direction, per ordered pair

(b) Panel means and contrasts

- **σ** symmetric similarity
- **ρ** directional part
- **ω** = log w, vulnerability

contrasts
source gap = σ part + ρ part; 2 × 2 cell contrasts; gradient cosine c tested against log τ (Mantel)

(c) Registered before measurement

panel family	K-a, K-b, K-c, CIFAR-10
two-by-two family	H1, H2, H3, CIFAR-10
attack side	best attack per source
masking check	AutoAttack
replication	K-a, K-b, K-c, CIFAR-100
gradient family	Mantel tests, all three
ImageNetV2 panel	descriptive, no decision

Stages ■ *[training: a, inputs]* ■ *[measurement: a, b]* ■ *[decision: c, registered]*